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Connects, holds, binds and supports other body tissues.

Consists of cells, fibers and ground substances.

Connective tissue types and subtypes are classified according to amounts, types and proportion of these components.

May be classified as:-

Embryonic connective tissue :-

a- Mesenchyme

b- Mucous connective tissue

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I. Embryonic connective tissue :-

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II. Adult connective tissue :-

A- Connective tissue proper

1-General C.T.

- Loose C.T.

- Dense C.T.

dense regular C.T

dense Irregular C.T.

2-Special C.T.

- adipose,

- reticular.

B-Supportive C.T.

- cartilage,

- bone,

C-Blood (highly specialized C.T)

Connective Tissue Fibers

There are three types of protein fibers in connective tissues:

collagen fibers, reticular fibers and elastic fibers.

- Collagen is the most abundant protein in the body % 30 of dry weight.

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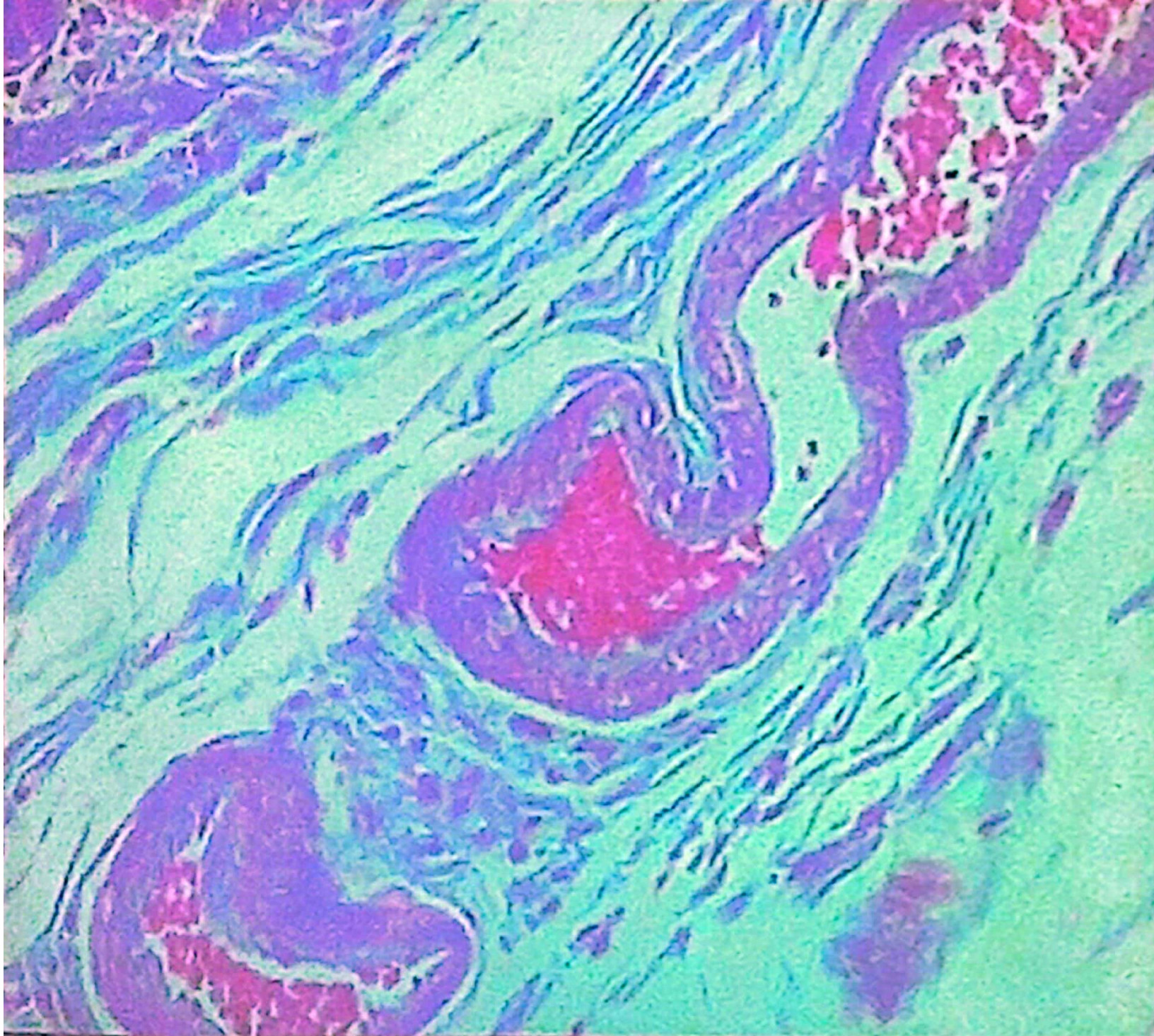
1. Collagen Fibers

• Collagen is the most abundant protein in the body. It is 33% dry weight.

Histological appearance

a- *Light microscopy (LM)*: Collagen occurring in large or small bundles of fibers or as individual fiber stains pink by (acidophilic) by H & E, it stains blue or green with trichrome stain.

b- *Electron microscopy (EM)*: All collagen fibrils have strips at intervals of 64 nm along their length. This periodicity reflects the staggering of tropocollagen molecules.



appearance LM

- Are similar to collagen fibers, but are thinner (0.1-0.5 μ m)
- By H & E are not visible

but can be demonstrated with periodic acid Schiff (PAS) reaction or with silver stains

- Are more highly glycosylated and form delicate silver-staining (argentaflin fibers) networks instead of thick bundles.

Structure

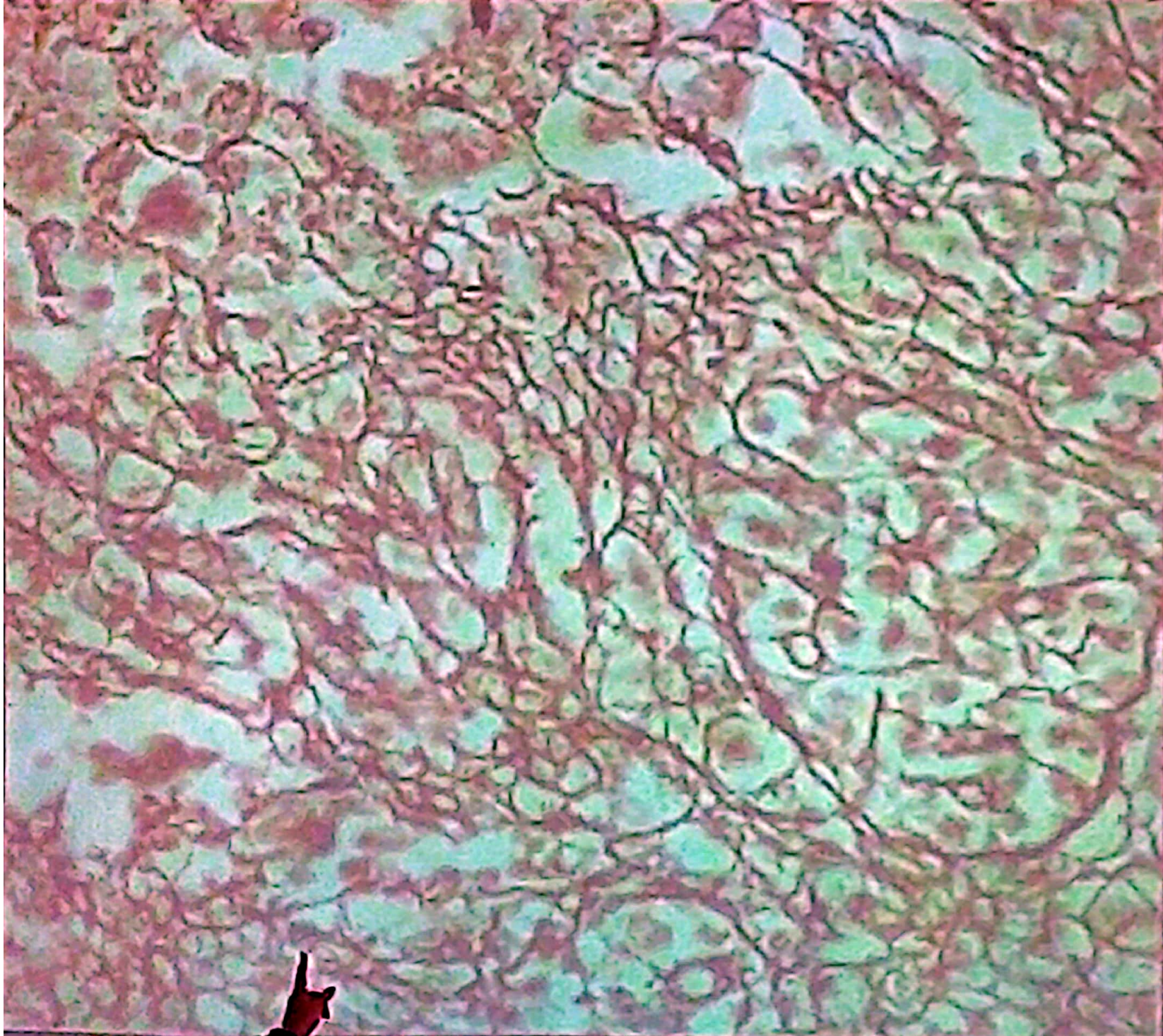
- Are composed loosely packed type III collagen and glycoproteins and proteoglycans.

Functions

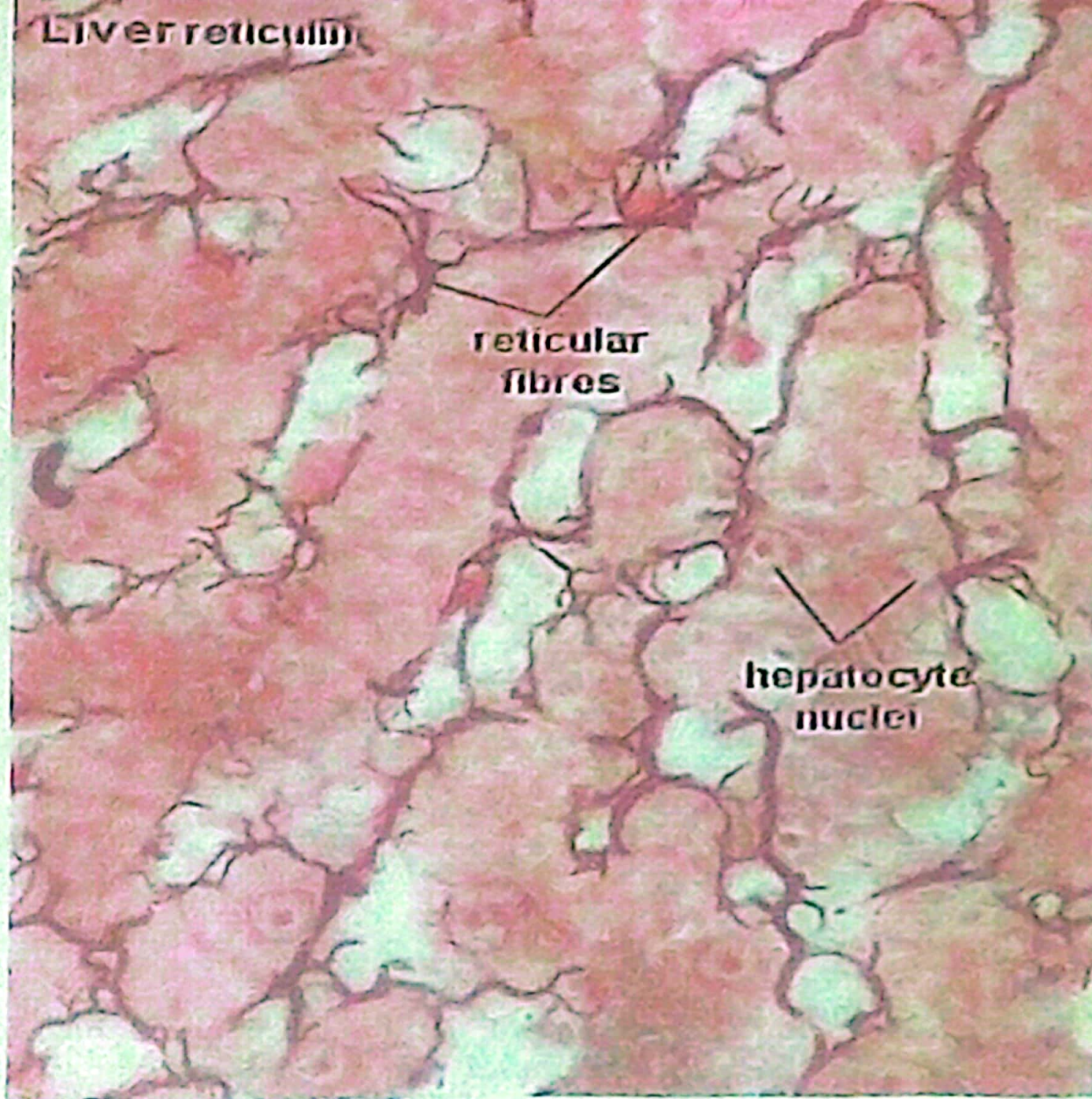
1-Support

- a -Blood—forming tissues (eg. Bone marrow, lymphatic organs) where they are produced by reticular cells
- b- Smooth muscle, where they are produced by smooth muscle cells
- c -Basement membranes, where they form the reticular layer

2-Tissue integrity



Liver reticulin



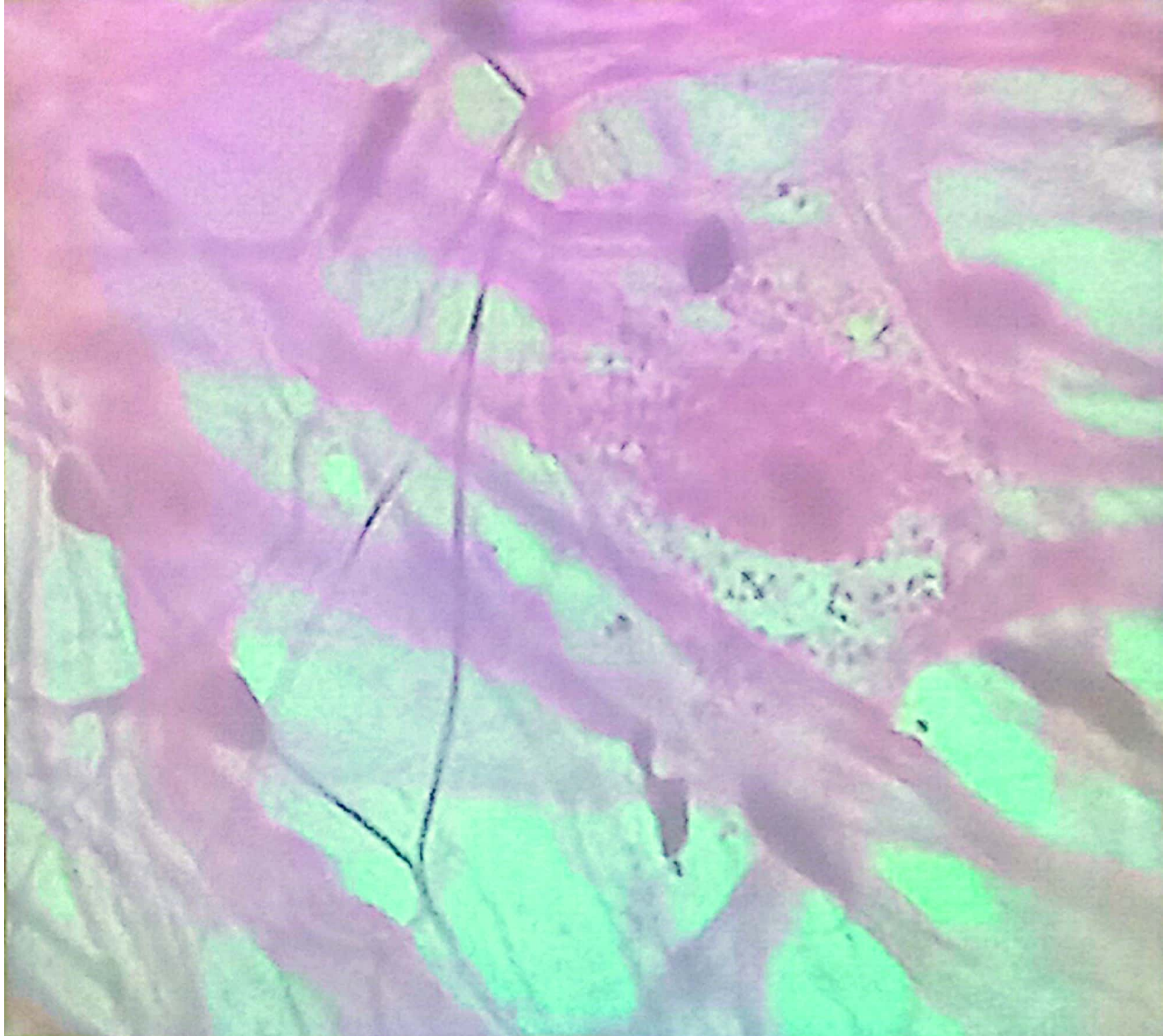
Components

a – *Proelastin*.

b- *Elastin*

LM appearance

- 1- Elastic fibers are acidophilic.
- 2- They are larger than individual collagen fibers and unlike collagen fibers, they are branched
- 3- it stained by Verhoeff's Elastic Stain , aldehyde fuchsin and Weigert's elastic stain





Ground substance

The ground substance consists mostly of glycoconjugates of proteoglycans and glycoproteins in addition to salts and tissue fluids. It cannot be seen by L.M.

1- Glycoprotein

2- Proteoglycans

Chondroitin sulfate occurs mainly in types of cartilage that are rich in collagen type II e.g. hyaline cartilage and elastic cartilage

Connective tissue cells can be grouped into 2 classes

1-fixed

2-wandering

1 -Fixed cells :Resident (native)to the tissue in which they are found. Most arise from mesenchyme, although some are immigrants from blood and are derived from hemopoietic stem cells.

a- Mesenchymal cell

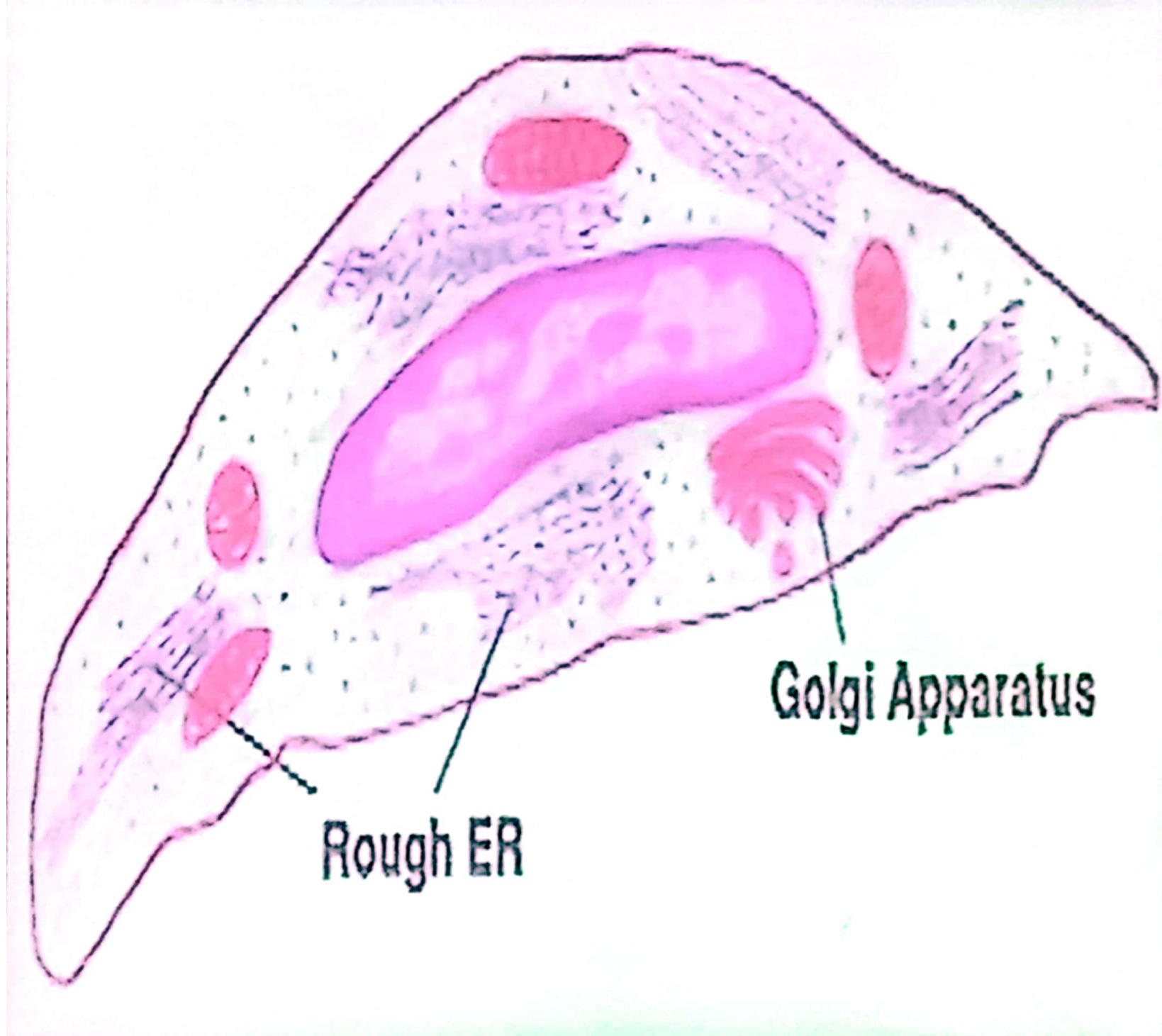
- Are precursors of most connective tissue cells.
- Mesenchymal cells have an irregular, star (stellate) shape with delicate branching cytoplasmic extensions which form an interlacing network. The oval nuclei have dispersed (extended) chromatin and visible nucleoli.
- Some mesenchymal cells remain undifferentiated in adult connective tissue (adventitial cells)= adult stem cell.

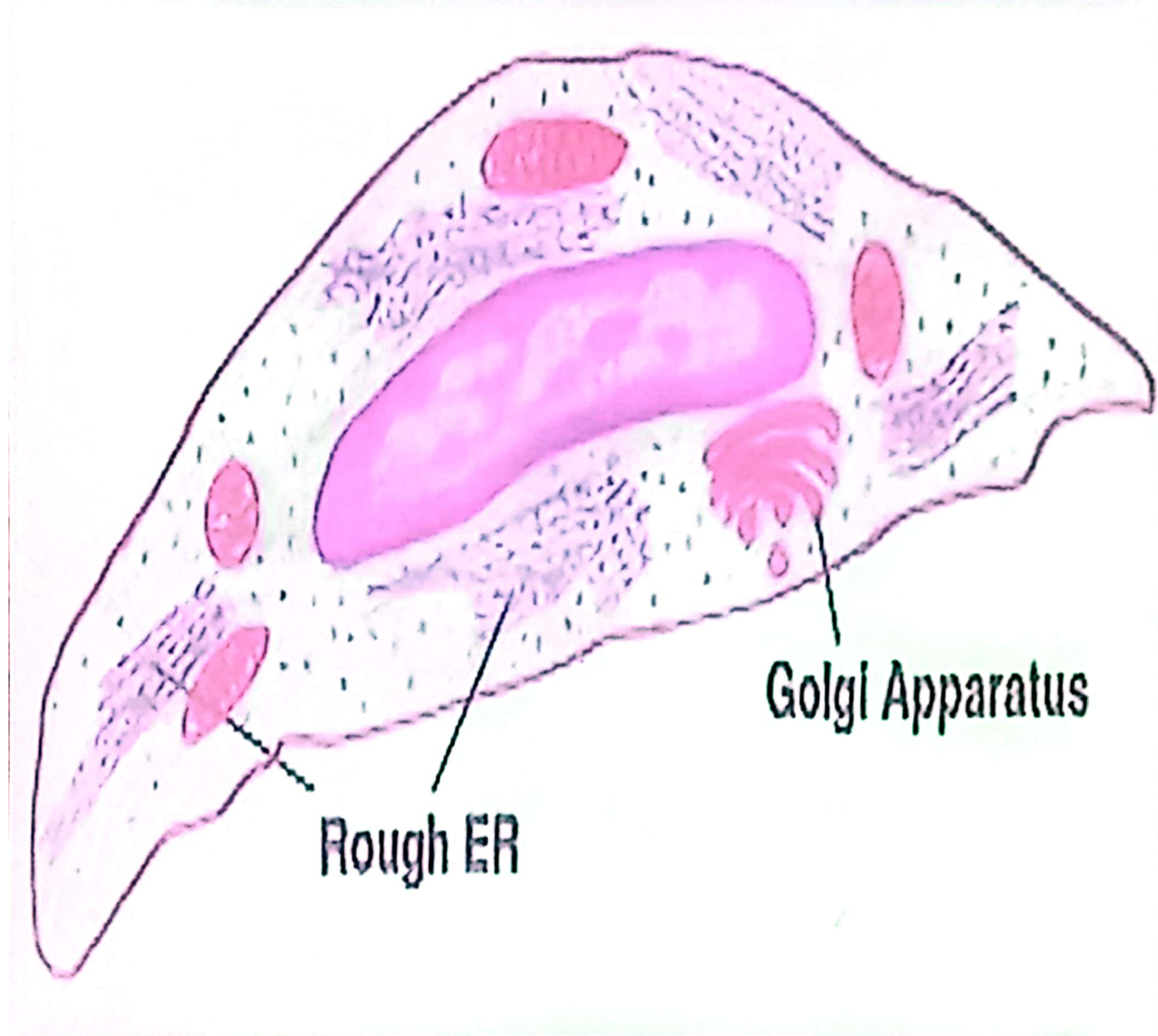
are spindle-shaped or branched cells that lie on the surface of capillaries and venules. They are a reserve population of relatively undifferentiated connective tissue cells that may differentiate into other connective tissue cells or into smooth muscle

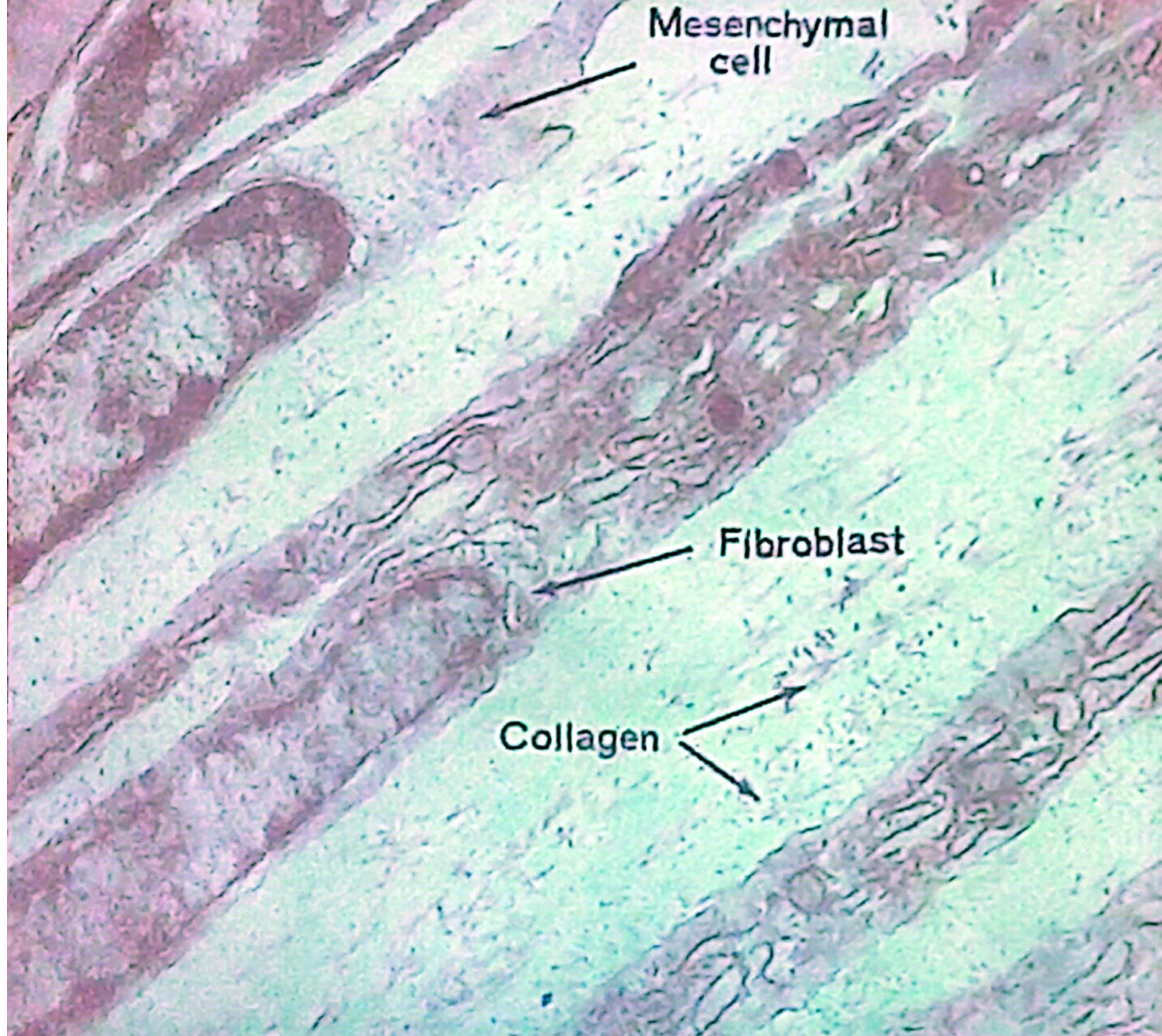
c- Fibroblasts (Active cell)

- Are the predominant and characteristic cells in connective tissue proper.
- They synthesize, secrete and maintain all the major components of the extra cellular matrix.
- The fibroblast resembles mesenchymal cells.
- They have large, ovoid, pale staining nuclei, their cytoplasm contains abundant rough endoplasmic reticulum (rER) and Golgi complexes.
- During growth and wound healing, activity is stimulated, the cells are enlarged, have more processes and the cytoplasm is basophilic.

- Fibrocytes - Are smaller and spindle-shaped with a dark elongated nucleus, fewer organelles and relatively inactive.







C- Reticular cells

Are stellate with long, thin cytoplasmic processes.

- Has a central, pale, irregularly rounded nucleus containing a prominent nucleolus.
- The cytoplasm containing numerous mitochondria, Golgi complex and rER.
- They produce reticular fibers.
- They are present only in reticular C.T

D- Adipose cells (adipocytes, fat cells)

(unilocular) White adipocyte

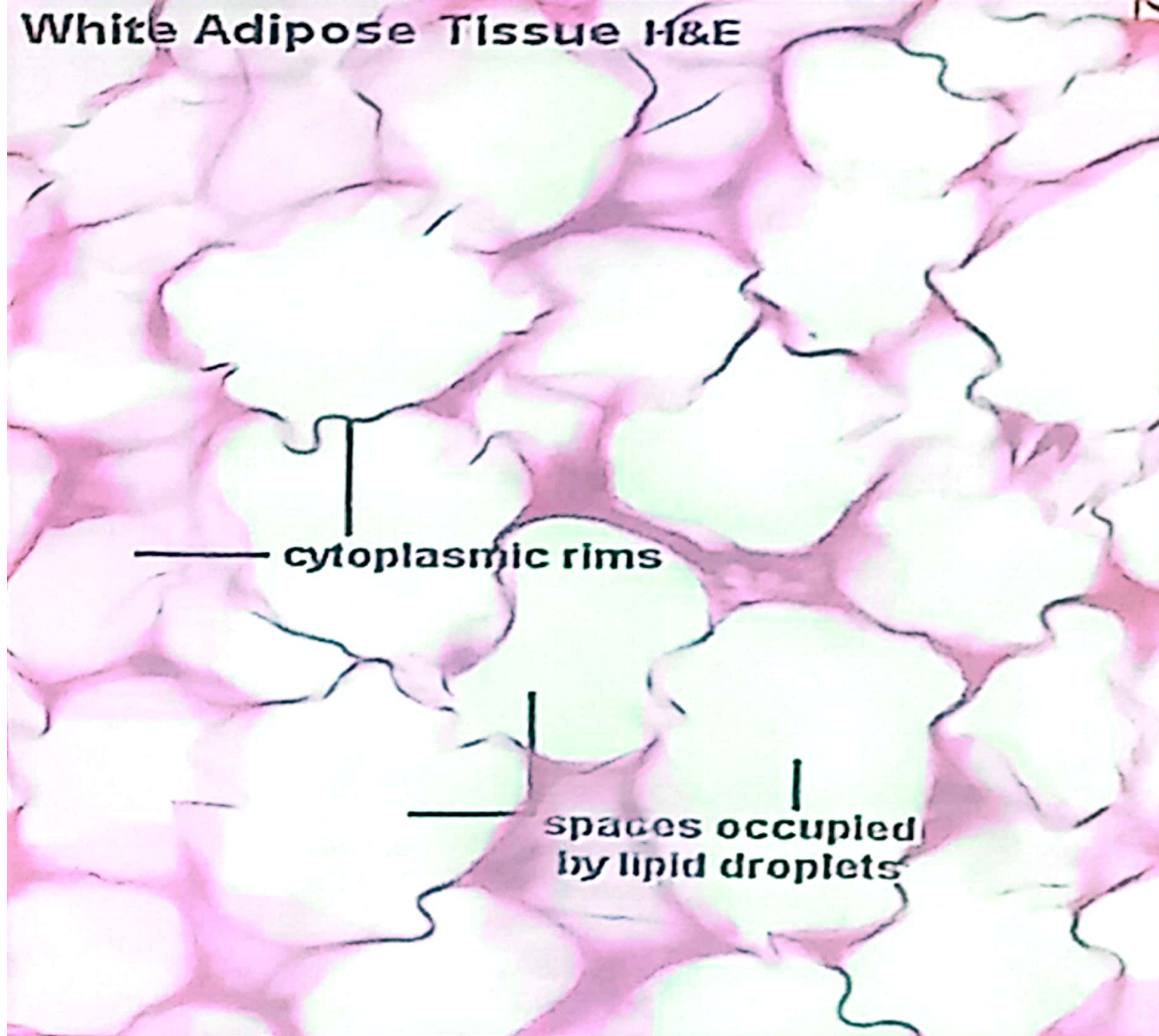
contains a single large fat droplet, displacing the heterochromatic nucleus and remaining cytoplasm to the cell periphery. Adipocytes in histological sections have a signet-ring appearance due to fat removal during preparation leaving only a flattened nucleus and a thin rim of cytoplasm. The perinuclear cytoplasm contains a Golgi complex, mitochondria, a small amount of rER and free ribosomes. The cytoplasm in the thin rim contains sER and pinocytotic vesicles.

Brown adipocyte (multilocular)

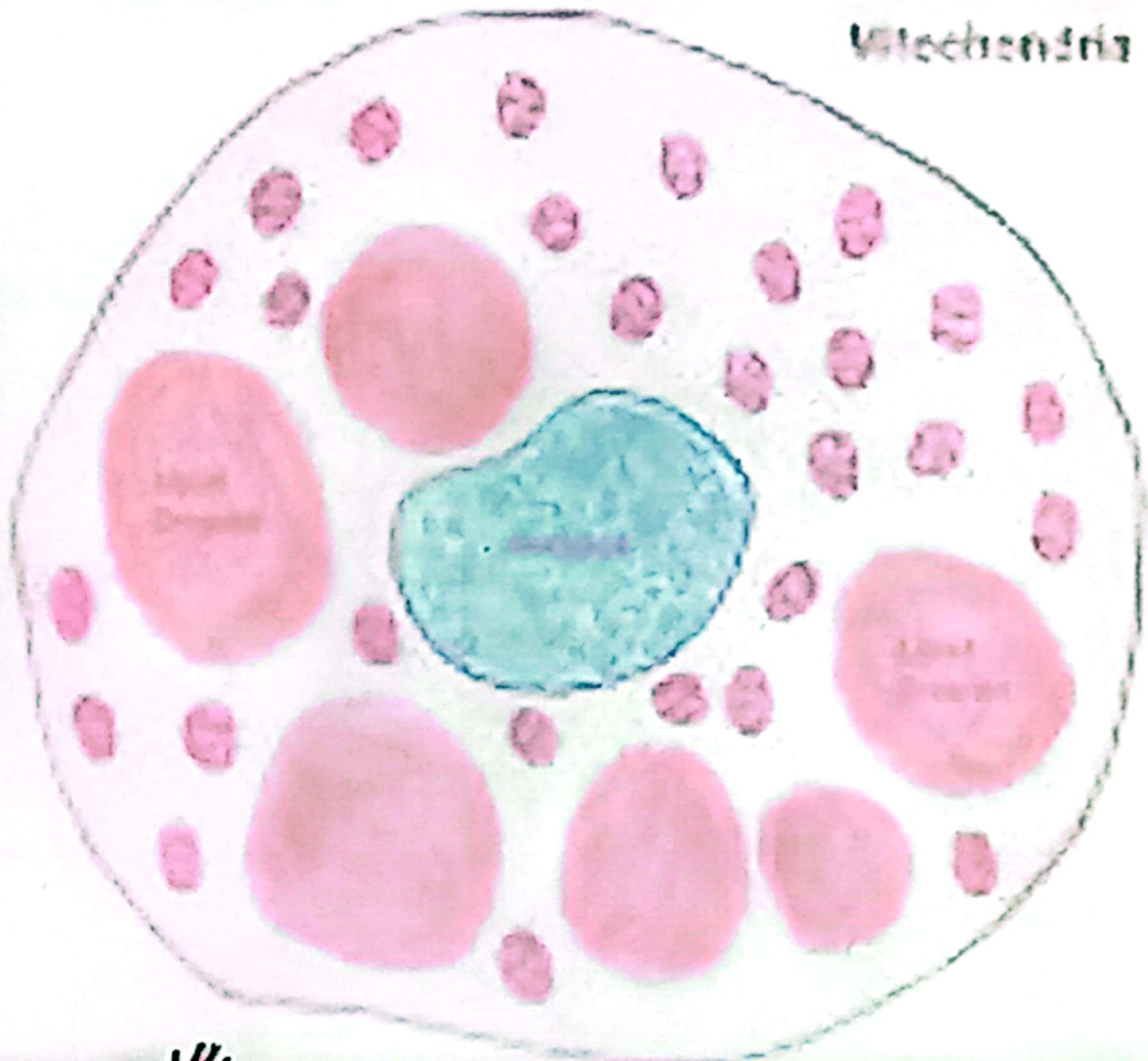
contains multiple small lipid droplets. Brown adipocytes are smaller than white adipocytes and have spherical, centrally located nucleus. They contain many mitochondria, the reddish brown colour is due to mitochondrial cytochromes.

Functions: Brown adipocytes metabolize their lipid stores to produce heat.

White Adipose Tissue H&E



Wolbachia



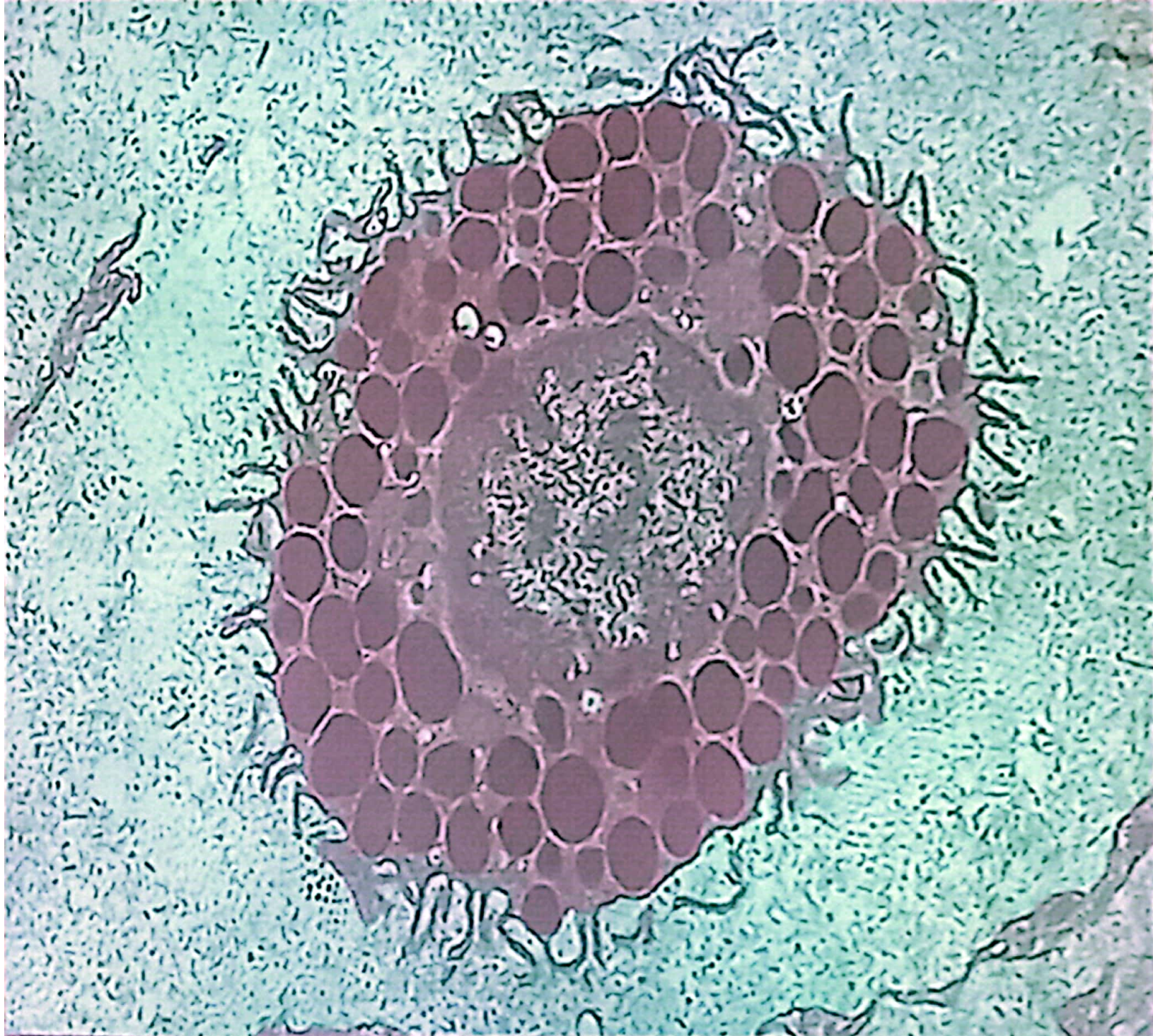
11.

2-Wandering cells(free cells)

a-Mast cells.

- Are located in loose connective tissue near small blood vessels.
- These are ovoid large cells (20-30 mm), derive from bone marrow.
- Are characterized by abundant basophilic cytoplasmic granules, water solubility of these granules leading to be lost during histological preparation.
- LM appearance:
 - a- Achromatic (non-staining) ; In most species the granules are unstained in routine preparation.
 - b- Metachromatic: Thus special techniques must be employed. When stained with certain basic dyes such as toluidine blue, the granules bind to the dye changing its colour to red, this property is known as *metachromasia*.
- It has been suggested that, polymerization of the dye molecule is responsible. The presence of macromolecules with electronegative radicals in the granules (heparin, polyanionic polymers) is thought to facilitate the polymerization leading to colour change.

features of mast cell a well developed Golgi complex and many small plasmalemma folds are present.



b- Macrophages (Histiocytes)

Are large cells derived from cells of the blood monocyte, that infiltrate the connective tissue and develop into phagocytes. Dye particles injected into the body are engulfed by these cells and accumulate their cytoplasmic granules. Macrophages contain many lysosomes and well developed Golgi complex. Macrophages may fuse to form multinuclear giant cell to remove large foreign objects such as splinters.

Structure of macrophages

They are irregular in outline, with an eccentric nucleus -
when they become actively phagocytic, they increase in size, extend
pseudopodia, and become more irregular in shape

Active cells have an extensive Golgi complex and many lysosomes. They
acquire cytoplasmic vacuoles and granules

Distribution: Most of the macrophage life span is spent in loose connective
tissue or reticular connective tissue, where they live for several months

The mononuclear phagocyte system (MPS) is composed of macrophages
throughout the body. In certain regions, macrophages have been given
special names e.g

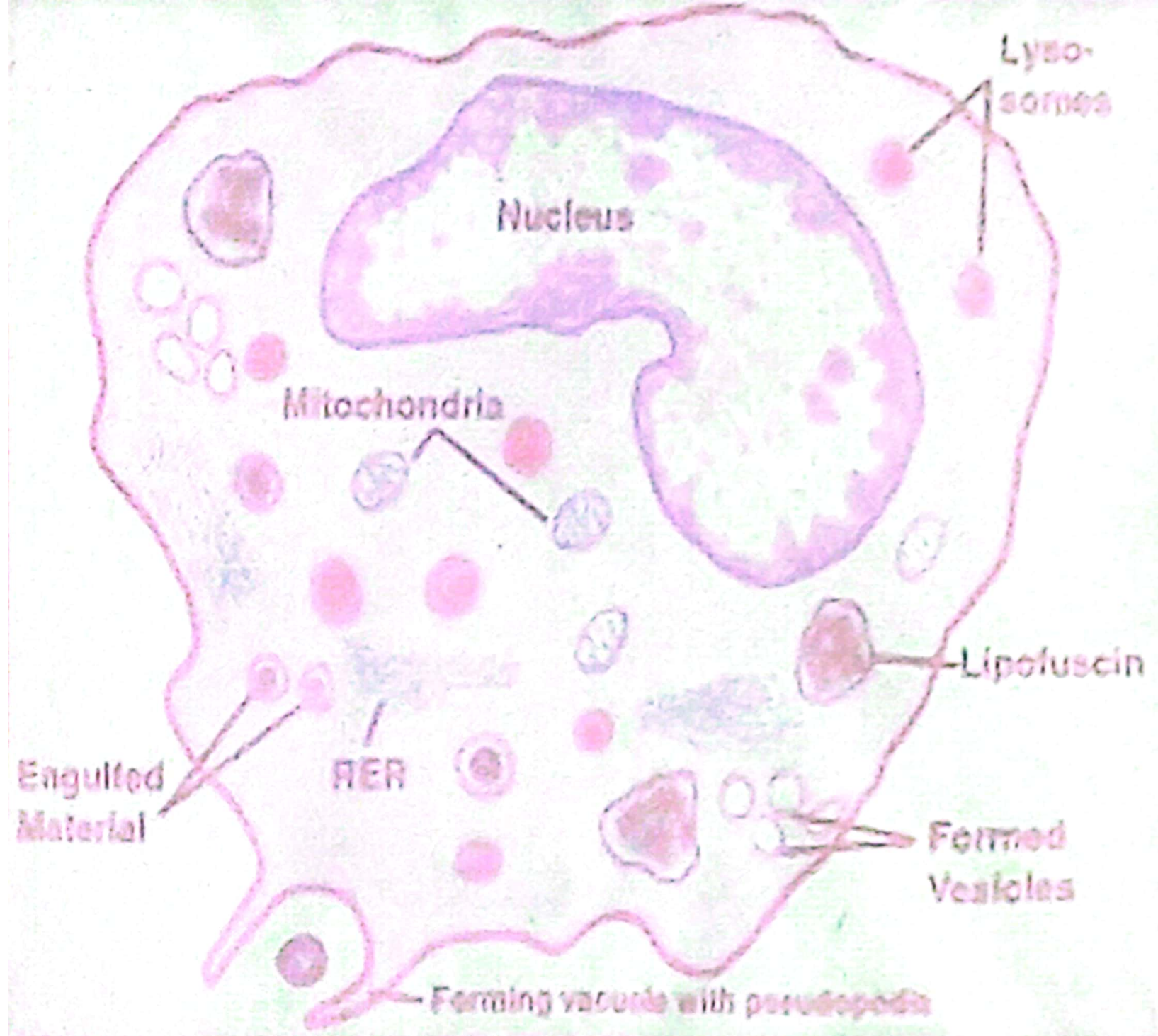
Kupffer cells in the liver

Alveolar macrophages Which can remove particles from the (dust cells):
mediated phagocytosis-lungs both nonselectively and by receptor

Osteoclasts in bone

Microglia in the central nervous system .

Function



c- Plasma cells

- Differentiate from antigen-stimulated B-lymphocytes. Structure plasma cells are large and ovoid, with an eccentric nucleus and abundant rER leading to basophilic cytoplasm.
- The characteristic "clock face" or "cart wheel" nucleus result from a large central nucleolus and several large condensed chromatin (heterochromatin) clumps regularly spaced around the inner surface of the nuclear envelope.
- These cells usually exhibit a clear juxtannuclear area (cytocenter) containing a well developed Golgi complex and centrioles.
- Distribution: Plasma cells are present in chronically inflamed tissue and in the mucous membranes of certain organ system.

• Connective tissue types:-

a- Mesenchyme

b- Mucous connective tissue

- **a- Mesenchyme** is the embryological tissue from which all connective tissue types are derived. It appears as loose array of stellate mesenchymal cells among them large intercellular spaces filled with ground substance (watery) and does not contain mature fibers.
- **b- Mucous connective tissue (Gelatinous connective tissue, Wharton's Jelly)** :is mainly composed of mucoid ground substances with few, fibroblasts and fibers distributed randomly in abundant ground substance (Jelly) are forming cushion to protect near by structures from pressure, e.g. umbilical cord, pulp of developing teeth and the comb of birds.

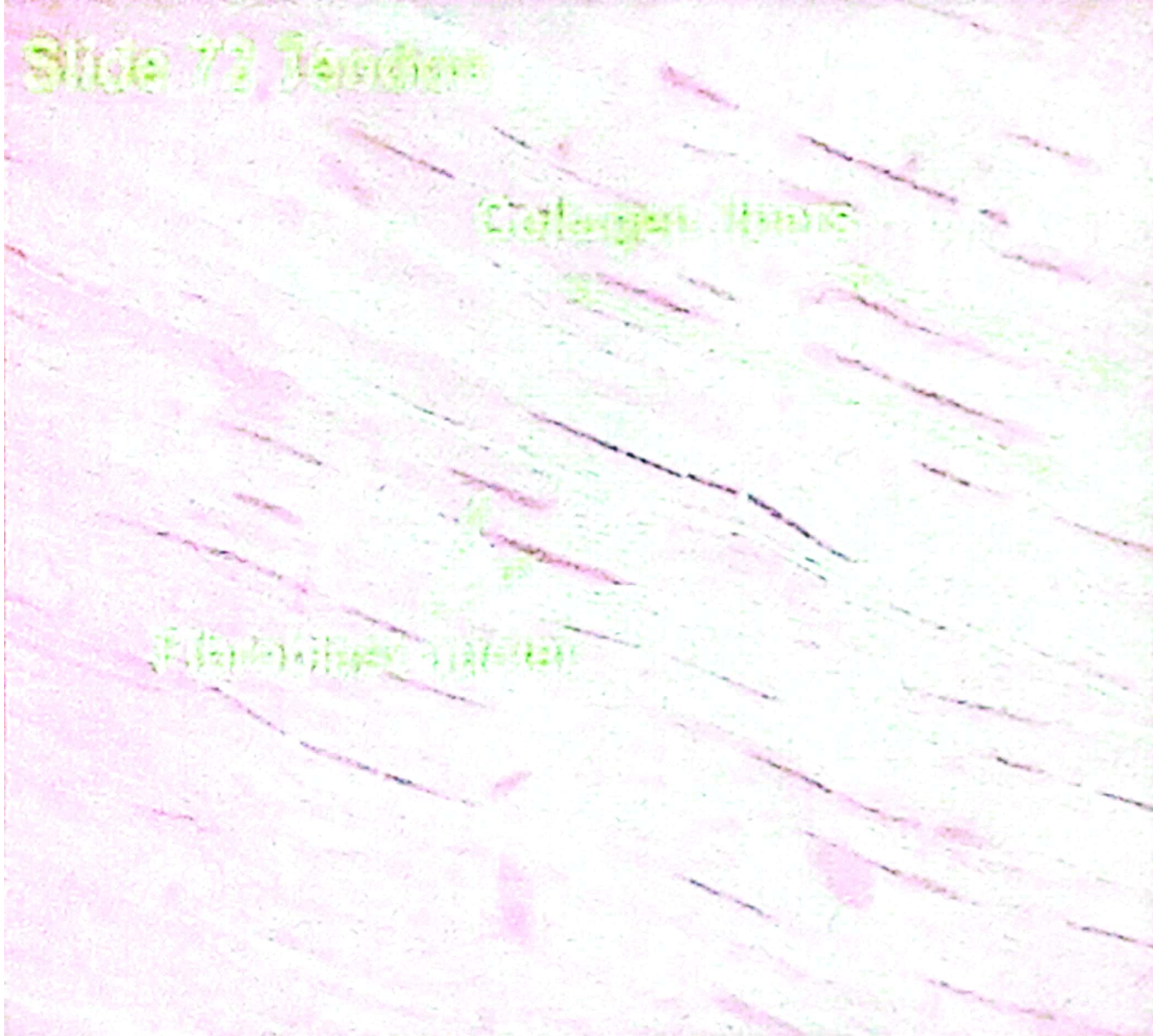
b- Dense connective tissue

- The fibers in dense connective tissue are more abundant than cells and amorphous ground substance.
- Collagen fibers are the predominant component of dense connective tissue (type I collagen) . The cells are predominantly mature fibroblasts. The ground substance is essentially identical to that of loose connective tissue but is less abundant. There are 2 types of dense connective tissue regular ,large collagen fiber bundles with a rope like arrangement while irregular with a fabric like arrangement.

Slide 72 Tendon

Collagen Bundle

Myofibril



Dense irregular c.t

- The components of this tissue are similar to those of regular one except the arrangement is interlacing that resists multidirectional mechanical stresses e.g. most organ capsules, dermis and mucocutaneous membrane.



the more abundant of 2 types, also termed *unilocular adipose tissue*.

General organization: Clusters of white adipocytes are divided into lobes and lobules by septa of loose connective tissue of variable density.

Individual cells are surrounded by a network of reticular fibers. The ground substance is sparse.

